

CHAPTER 2

PLANKTON AND PLANKTON COMMUNITIES

CHAPTER OUTLINE AND SIGNIFICANT ELEMENTS

This long chapter is divided into nine sections, but a brief preliminary unit entitled DEFINITIONS begins the chapter. That unit is designed to acquaint the student with the size range of organisms that compose the plankton. The first three sections thereafter treat THE PHYTOPLANKTON, THE VIRUSES, and THE ZOOPLANKTON, respectively, while a follow-up section discusses FLOTATION MECHANISMS of these small organisms. The next three sections cover various aspects of primary production: PRIMARY PRODUCTION, FACTORS AFFECTING PRIMARY PRODUCTIVITY, and PRIMARY PRODUCTIVITY OF THE BIOSPHERE. The final two sections of this chapter discuss ecological interactions among the plankton: THE OCEAN ECOSYSTEM: THE CLASSICAL MODEL and THE OCEAN ECOSYSTEM: A CHANGING MODEL.

Definitions

bacterioplankton	mesoplankton	picoplankton
femtoplankton	microplankton	plankton
holoplankton	nanoplankton	viroplankton
macroplankton	nekton	zooplankton
megaplankton	net plankton	
meroplankton	phytoplankton	

I. The Phytoplankton

This short section provides a précis on the important phytoplankton taxa. Two groups are recognized, based on size: Net phytoplankton are captured by fine-meshed plankton nets ($\geq 25 \mu\text{m}$), while a group of small, poorly understood forms passes through these nets. This part of the chapter is divided into three sections:

- Diatoms
- Dinoflagellates
- The Smaller Phytoplankton

alpha proteobacteria	domoic acid	prochlorophytes
auxospore	epitheca	prokaryote (cf. eukaryote)
bioluminescence	frustule	proteorhodopsin
blooms	green algae	Red Sea
blue-green bacteria	haptonema	red tide
cellulose	haptophytes	silicoflagellates
coccolithophores	hypotheca	silicon dioxide
cryptomonads	marine snow	theca
cyanobacteria	nitrogen fixation	<i>Trichodesmium</i>
diatom valves (theca)	<i>Noctiluca</i>	zooxanthellae
diatoms	particulate organic	
dinoflagellates	carbon (POC)	

1. Most diatoms (a dominant form of the net phytoplankton) construct a silicon shell, or frustule, that resembles a circular petri dish, with a top and a bottom. Other diatoms are not circular; they are more or less elongate (e.g., pennates). The way in which diatoms reproduce themselves (Fig. 2.1) makes for a minor, but interesting, variation of the binary-fission-growth model: 1, 2, 4, 8, . . . , auxospore,

12 Instructor's Manual for Marine Biology

The ability of diatoms to produce domoic acid, a natural amino acid with toxic properties that can cause amnesic shellfish poisoning (ASP), provides an additional incentive for their study. This toxin accumulates in many commercially important shellfish. When these shellfish are eaten by humans, the toxin acts as a neuronal depolarizer and may cause depolarization of the hippocampus. Consult the following Web sites for additional information (N.B.: All listed sites were active as of March 26, 2004):

- (a) NOAA's Northwest Fisheries Science Center, West Coast Marine Biotoxins & Harmful Algal Blooms Newsletter: <http://www.nwfsc.noaa.gov/hab/index.htm>;
- (b) Washington Department of Fish and Wildlife: <http://www.wdfw.wa.gov/fish/shelfish/razorclm/domacid.htm>;
- (c) Canadian Food Inspection Agency Food Facts at: <http://www.cfia-acia.agr.ca/english/corpaffr/foodfacts/aspdae.shtml>.

2. Dinoflagellates (members of the net phytoplankton) are important for five reasons:

- (a) They make an important contribution to community primary production. (Some are not photosynthetic; these forms graze to acquire their energy.)
- (b) They are often armored in cellulose and produce long spines. (Spines are a recurrent theme in planktonic organisms, often being used as antipredator devices.)
- (c) Some are capable of producing toxins that may cause the death of marine animals under certain conditions—for example, red tide.
- (d) Several species are capable of bioluminescence.
- (e) The nonmotile zooxanthellae stage is an endosymbiont in the tissues of certain invertebrates.

A. PHYTOTOXIN PRODUCTION: Toxins produced by dinoflagellates include okadaic acid, ciguatoxins, saxitoxins, and brevetoxins. Domoic acid (see item 1 in this section) is produced by diatoms of the genus *Pseudonitzschia*.

Saxitoxins are water-soluble, nondepolarizing tetrahydropurines that cause paralytic shellfish poisoning (PSP). There are at least 12 compounds composing this class of toxins. As Na^+ -channel blockers, they inhibit conduction of the peripheral nerves, leading to hypotension, lowered respiration, and respiratory paralysis.

Brevetoxins are lipid-soluble, depolarizing neurotoxins that cause neurotoxic shellfish poisoning (NSP). As Na^+ channel activators, they cause nausea, vomiting, and diarrhea. Although less lethal than saxitoxin, brevetoxins can irritate the respiratory tract if the toxins are breathed in with sea spray!

Ciguatoxins are lipid-soluble toxins with effects that are similar to those caused by brevetoxins. They also are Na^+ -channel activators. These toxins are the agent responsible for ciguatera fish poisoning.

Okadaic acid and other lipid-soluble toxins are phosphatase inhibitors. These toxins affect cell-level metabolic processes directly.

***Pfiesteria* toxin** has an unknown mode of action, and it has not yet been characterized biochemically. [N.B.: Some workers have questioned the existence of this toxin.] Its effects on the human nervous system are severe in the extreme; yet it does not have significant effects on certain marine invertebrates, such as the copepod *Acartia* and the rotifer *Brachionus*. Following are some selected references:

Anderson, D.M. 1994. Red tides. *Sci. Am.* 271(2): 62–68.

Burkholder, J.M. 1999. The lurking perils of *Pfiesteria*. *Sci. Am.* 281(2): 42–49.

Burkholder, J.M., and H.B. Glasgow, Jr. 1997. *Pfiesteria piscicida* and other *Pfiesteria*-like dinoflagellates: behavior, impacts, and environmental controls. *Limnol. Oceanogr.* 42: 1052–1075.

Glasgow, H.B., Jr., et al. 1995. Insidious effects of a toxic dinoflagellate on fish survival and human health. *J. Toxicol. Environ. Health* 46: 501–522.

Kviteck, R.G., A.R. DeGange, and M.K. Beitler. 1991. Paralytic shellfish poisoning toxins mediate feeding behavior in sea otters. *Limnol. Oceanogr.* 36: 393–404.

Matter, A.L. 1994. Paralytic shellfish poisoning: toxin accumulation in the marine food web, with emphasis on predatory snails. EPA Region 10, Seattle, WA. 44 pp. EPA 910/R-94-005.

Miller, T.R., and R. Belas. 2003. *Pfiesteria piscicida*, *P. shumwayae*, and other *Pfiesteria*-like dinoflagellates. *Res. Microbiol.* 154(2): 85–90.

Samet, J., et al. 2001. *Pfiesteria*: review of the science and identification of research gaps. Report for the National Center for Environmental Health, Centers for Disease Control and Prevention. *Environ. Health Perspect.* 109 (Suppl 5): 639–659.

Scope, G.G. 1989. Marine envenomations and aquatic dermatology. *Am. Fam. Phys.* 40(2): 97–106.

Todd, E.C.D. 1993. Domoic acid and amnesic shellfish poisoning. *J. Food Prot.* 56: 69–83.

Tuner, J.T., and P.A. Tester. 2000. Toxic marine phytoplankton, zooplankton grazers, and pelagic food webs. *Limnol. Oceanogr.* 42: 1203–1214.

- B. BIOLUMINESCENCE:** Some dinoflagellates are capable of bioluminescence. Exactly why they exhibit this feature has been of interest to students of the marine plankton for some time. An interesting twist, of course, is that some bacteria are also bioluminescent and have coevolved with certain invertebrates such as squid. Some references are as follows:

Abrahams, M.V., and L.D. Townsend. 1993. Bioluminescence in dinoflagellates: a test of the burglar alarm hypothesis. *Ecology* 74: 258–260.

Esaias, W.E., and H.C. Curl, Jr. 1972. Effect of dinoflagellate bioluminescence on copepod ingestion rates. *Limnol. Oceanogr.* 17: 901–906.

Glausiusz, J. 1997. The importance of being infected. *Discover* 18: 30.

McFall-Ngai, M. 1990. The anatomy and morphology of the adult bacterial light organ of *Euprymna scolopes* Berry (Cephalopoda: Sepiolidae). *Bio. Bull.* 179: 332–339.

Toner, M. 1994. When squid shine . . . and mushrooms glow, fish twinkle and worms turn into stars. *International Wildlife* 24: 30–37.

3. Several small and lesser known phytoplankton make significant contributions to the primary production of the oceans (for example, coccolithophores).
4. A clear differentiation should be made between the eukaryotic phytoplankton and the prokaryotes (that is, bacteria and cyanobacteria) in terms of their cellular ultrastructure and biochemistry. Bacteria will colonize and utilize, as a carbon/energy (C/E) source, the gelatinous material in the water column known as marine snow.
5. Phytoplankton are those organisms capable of carrying on photosynthesis. Because of their small size and rapid growth capabilities, they can quickly develop a bloom, or population explosion. If we assume no cell death, the mathematics of this bloom is simple and may be illustrated as follows:

cell numbers	1	→	2	→	4	→	8	→	16	→	32 . . .
generation			1		2		3		4		5

OR

Number of algae = 2^n , where n = the number of generations.

II. The Viruses

This brief section introduces students to viruses, which at 10^{10} viruses per liter are the most numerous biotic entities of the seas. For additional general information, consult any good microbiology textbook.

DNA viruses

intracellular parasites

RNA viruses

14 Instructor's Manual for Marine Biology

- Dominique, M., and C. Brussaard. 1999. Enumeration of marine viruses in culture and natural samples by flow cytometry. *App. Environ. Microbio.* 65: 45–53.
- Mei, M.L., and R. Danovaro. 2004. Virus production and life strategies in aquatic sediments. *Limnol. Oceanogr.* 49: 459–470.
- Wommack, K.E., and R.R. Colwell. 2000. Virioplankton: viruses in aquatic ecosystems. *Microbiol. Mol. Biol. Rev.* 64: 69–114.
- Zhong, Y., et al. 2002. Phylogenetic diversity of marine cyanophage isolates and natural virus communities as revealed by sequences of viral capsid assembly protein gene g20. *App. Environ. Microbiol.* 68: 1576–1585.

III. The Zooplankton

The zooplankton appear to be more diverse than the phytoplankton, with many holoplanktonic as well as meroplanktonic species. Like the phytoplankton, the zooplankton are separated into distinct size categories. In this chapter, we discuss the zooplankton in two subsections:

- Copepods
- Other zooplankton

Amphipoda	Decapoda	net zooplankton
Annelida	Euphausiacea	Ostracoda
arrowworms	Foraminifera (forams)	picozooplankton
Arthropoda	foraminiferan ooze	Polychaeta
bubble raft	Gastropoda	Protista
calcium carbonate	geodesic dome	pteropods
Chaetognatha	heteropods	Radiolaria
Chordata	holozooplankton	radiolarian ooze
Ciliophora	Hydrozoa	Rotifera
Cladocera	jellyfish	salps
Cnidaria	Larvacea	Sarcomastigophora
Copepoda	Mollusca	Scyphozoa
copepodid	Mysidacea	silicon dioxide
Crustacea	nanozooplankton	Siphonophores
ctenes	nauplius	squids
Ctenophora	Nemertea	Thaliacea

1. Copepods, at all stages of their life history, are often the most abundant members of the zooplankton. The generalized life cycle of copepods is as follows: nauplius larva, with six naupliar stages indicated by NI–NVI, and copepodid, again with six stages, indicated by CI–CVI. The adult stage is the last copepodid stage (CVI).

Dussart, B.H., and D. Defaye. 1995. Introduction to the Copepoda. Guides to the identification of the microinvertebrates of the continental waters of the world, Volume 7. The Hague: SPB Academic Publ. [This publication is about freshwater copepods, but will serve to illustrate the concepts.]

Hicks, G.R.F., and B.C. Coull. 1983. The ecology of marine meiobenthic harpacticoid copepods. *Ann. Rev. Oceanogr. Mar. Bio.* 21: 67–175.

Ikeda, T., et al. 2001. Metabolic rates of epipelagic marine copepods as a function of body mass and temperature. *Mar. Bio.* 139: 587–597.

Mauchline, J. 1998. The biology of calanoid copepods. *Advances in Marine Biology*, Volume 33. San Diego: Academic Press.

Mazzocchi, M.G., et al. 1995. *Atlas of marine zooplankton. Straits of Magellan: Copepods*. New York: Springer-Verlag.

Williamson, C.E. 1991. Copepoda, in J. Thorp and A. Covich, eds. *Ecology and classifications of North American freshwater invertebrates*. San Diego: Academic Press. [This publication also is about freshwater copepods.]

2. Other important zooplankton taxa include the foraminiferans (forams), radiolarians, cnidarians, ctenophores, and the larval stages of many phyla. The skeletons—or tests—of foraminiferans (CaCO_3) and radiolarians (SiO_2) can dominate sediments, forming deep layers of ooze. Depending on the dominant form composing this ooze, it may be called radiolarian or foraminiferan ooze. The geodesic dome clearly evolved in the radiolarians (and, depending on your interpretation, perhaps the forams) long before the architectural advancement of Buckminster Fuller. While buckminsterfullerene (buckyballs) has been widely publicized as a marvelous invention, recent reports indicate that it, too, is an ancient structure.
3. The meroplankton are, as a group, more taxonomically diverse than the holoplankton.
4. Prior to the research of Rudi Strickler and his coworkers (for example, Koehl and Strickler, 1981), it was thought that copepods fed by creating circular feeding currents and that this action led to a reprocessing of filtered water in twin vortices, one on each side of the animal. This incorrect view is still found in some textbooks. Strickler and his coworkers demonstrated that the water is processed only once as it passes by the animal. The older model was based on observations of water currents produced by copepods held in small depression slides, while the newer model tracked animals in much larger volumes of water (ca. $10^6\times$). See the Enrichment Exercises at the end of this chapter for additional information on this topic.
5. A better understanding of the distribution of zooplankton is now possible through the use of advanced electronic systems.

Davis, C.S., S.M. Gallager, and A.R. Solow. 1992. Microaggregations of oceanic plankton observed by towed video microscopy. *Science* 257: 230–232.

Everson, I. 1990. Implications of a new acoustic target strength for abundance estimates of Antarctic krill. *Nature* 345: 338–340.

Koehl, M.A.R., and J.R. Strickler. 1981. Copepod feeding currents: Food capture at low Reynolds number. *Limnol. Oceanogr.* 26: 1062–1073.

Ricketts, C., et al. 1992. An assessment of the biological and acoustical characteristics of swarms of Antarctic krill. *Deep Sea Res. (Part A)* 39: 359–371.

IV. Flotation Mechanisms

This section discusses the phenomenon of plankton flotation in four subsections:

- Principles
- Reduction of Overweight
- Changes in Surface Resistance
- Water Movements

convection cells

density

gas storage (floats)

ion replacement

Langmuir convention cells

lipid storage

Noctiluca

oil drops

osmosis

Physalia

sinking rate

specific gravity

spines

Stoke's law

surface resistance

viscosity

1. Planktonic organisms tend to sink and, being weak swimmers, have a problem remaining in the plankton. The rate of sinking of small particles may be studied by employing Stoke's law, which is given in one of its forms in the text. Several adaptations have been identified that reduce sinking of both phyto- and zooplankton. (See below.)
2. Sinking of plankton in the open water is a serious problem, and several features of these small organisms seem to be adapted to overcoming the effects of gravity. Heavier chemical ions may be replaced with lighter ones (ammonium ions), or gas can be produced and stored in chambers. Certain food reserves (fats and oils) also provide increased buoyancy, but there is a cost to lipid storage: visibility to zooplanktivorous fish that feed by sight. Increased surface area relative to volume also increases resistance to sinking.

16 Instructor's Manual for Marine Biology

3. Some water movements, particularly Langmuir convection cells, can help keep plankton in the upper waters. These currents tend to collect floating cells and debris (trash, tar balls, oil, etc.) in the convergence region. The result is windrows, a series of parallel lines of floating stuff several meters apart, but stretching on for hundreds or thousands of meters.

V. Primary Production

Three units cover this section on marine carbon fixation and primary production:

- Primary Productivity
- Measurement of Primary Productivity
- Standing Crop

¹⁴ C method	glass-fiber filters	photorespiration
APAR	gross 1° production	photosynthesis
ATP assay	light/dark bottle method	photosynthetically
biomass	light-dependent reactions	active solar
bottle effect	light-independent reactions	radiation (APAR)
Calvin–Benson cycle	membrane filters	primary productivity
community production	net 1° productivity	satellite imaging
community respiration	net community	standing crop
compensation depth	photosynthesis	transparency
compensation intensity	net production	utilization efficiency (<i>e</i>)
critical depth	new production	Winkler O ₂ method
flow cytometer	PAM fluorometer	
FRR fluorometer	particle counters	

1. All students should be familiar with the summary equation of photosynthesis (that is, the light and dark reactions, or rather the light-dependent and light-independent reactions, respectively). The light-dependent reactions produce ATP and NADPH + H⁺; the light-independent reactions (= Calvin–Benson Cycle) use ATP and NADPH + H⁺ to fix CO₂ into hexose sugars:



2. The term “production” is used to quantify the amount of carbon locked into biomass (that is, grams of carbon per unit area or volume). In this context, it refers to the amount of carbon that has been fixed by photosynthesis (PS). The term “productivity” denotes a rate function: amount of carbon fixed per unit area (or volume) per unit time. Production and productivity can refer to the total amount of PS-fixed carbon (gross production, or GP) or the net amount (net production, or NP) that remains when respiration (*R*) is subtracted from GP (GP – *R*).
3. The compensation depth (PS = *R*) varies with the amount of light reaching into the water column. Therefore, the compensation intensity may be a simpler concept for students to appreciate. Both of these measures will vary with the composition of the phytoplankton. The compensation depth is generally assumed to be approximated by the depth at which 1% of the incident surface illumination (*I*₀) is reached.
4. While replete with sources of error, the light and dark bottle techniques, using either the oxygen method or the ¹⁴C method, are important for a variety of reasons: They represent classical measurements; they can be useful independent tests of more sophisticated techniques; they may be implemented with limited equipment.
5. Advanced techniques using particle counters, flow cytometry, ATP assays, and satellite imaging require sophisticated laboratories, but incredible advances are being made by the application of these new techniques. Examples of two papers on these techniques are as follows:

Cavender-Bares, K.K. 1998. A dual sheath flow cytometer for shipboard analyses of phytoplankton communities from the oligotrophic oceans. *Limnol. Oceanogr.* 43: 1383–1388.

Li, W.K.W. 1997. Cytometric diversity in marine ultraphytoplankton. *Limnol. Oceanogr.* 42: 874–880.

VI. Factors Affecting Primary Productivity

This long section is divided into three parts:

- Physical and Chemical Factors
- Geographical Variations in Productivity
- Productivity in Inshore and Coastal Waters

absorption	limiting factors	thermocline
compensation depth	mixing	trace metal
critical depth	photic zone	transparency
extinction coefficient (<i>k</i>)	production	turbulence
fall bloom (pulse)	scattering	upwelling
hydrography	spring bloom	
<i>I_z</i> and <i>I₀</i>	thermal stratification	

1. The same factors important to plant growth in terrestrial systems are important to plant growth in aquatic systems. Two limiting factors are critical: light and nutrient concentration. Hydrography is a term that comprises all the factors important in controlling phytoplankton production.
2. Factors influencing the quality and quantity of light energy reaching phytoplankton include absorption of light energy by water, dissolved and suspended materials, wavelength, the amount of light reaching the water surface, and surface conditions. These factors are integrated in the following equation, which may be meaningless to many students:

$$I_z = I_0e^{-kz}$$

However, to start the class on a discussion of the transmission of light through water, I employ the following exercise: Take two pieces of chalk, one large and the other much smaller. Place the larger piece in a flask of water and hold it up for the class to see. Now grind the other piece up, using a mortar and pestle, and suspend the powder in a second flask of water. Obviously, the contents of the second flask will be cloudy, and less light will pass through the second flask than the first. The more chalk dust, the less light will pass through the flask. After this simple demonstration, students are usually able to deduce that the decrease in light is a function of particle size. Using Spectronic 20s, students can then test their hypothesis in a more rigorous way. (N.B.: I am indebted to W.T. Edmondson, who first illustrated this simple teaching tool for me.)

3. The major nutrients required by phytoplankton include nitrogen compounds (nitrate and nitrite), phosphate, and silicon:



Recent research indicates that iron (Fe) also may be limiting in the open ocean. Using that information, some workers have proposed that massive fertilization schemes be adopted to enrich parts of the southern oceans with Fe. This fertilization would presumably increase their productivity and, in the process, increase their capacity to act as sinks for elevated atmospheric CO₂ (a greenhouse gas). See also the discussion of the IronEx experiments in Chapter 11.

Coale, K.H., and K.S. Johnson. 1998. IronEx-I, an in situ iron-enrichment experiment: Experimental design, implementation and results. *Deep-Sea Research Part II, Topical Studies in Oceanography* 45: 919–946.

Cullen, J.J. 1995. Status of the iron hypothesis after the Open-Ocean Enrichment Experiment. *Limnol. Oceanogr.* 40: 1336–1343.

Martin, J.H. 1994. Testing the iron hypothesis in ecosystems of the equatorial Pacific Ocean. *Nature* 371: 123–129.

4. The concepts of compensation depth and critical depth may be confusing to students. Compensation depth is the depth of which the rate of photosynthesis equals the rate of respiration. This concept is straightforward and is represented graphically in Fig. 2.37 in the textbook by the point where

18 Instructor's Manual for Marine Biology

the curved line (indicating the photosynthetic rate) crosses the vertical straight line (indicating the respiration rate).

Critical depth is the depth at which the total phytoplankton biomass may be circulated and still spend enough time above the compensation depth to have a total production that is greater than or equal to its total respiration. In Fig. 2.37 in the textbook, the area under the curve of the photosynthetic rate, but above the line of the respiration rate, is net production. One may consider this quantity to be an energy reserve or positive balance that may be applied against additional respiration accrued if the population circulates below the compensation depth (sort of like saving for the proverbial rainy day).

5. The factors influencing the pattern of primary production (spring bloom, lower summer production, and fall pulse) are complex, and lectures on them can degenerate into considerable hand waving and other histrionics to explain the phenomenon. Nevertheless, this topic makes for a good show if presented correctly. Inshore production is influenced considerably by runoff from the land: Nutrient levels are much higher, as is the sediment load. A persistent thermocline is often not present. Human activity exacerbates these factors by adding more nutrients (eutrophication) and toxic compounds (pollution).

Livingston, R.J. 2000. *Eutrophication processes in coastal systems*. Boca Raton, FL: CRC Press.

6. Two terms used to discuss controlling features of phytoplankton (and zooplankton) population dynamics are "bottom-up processes" and "top-down processes." That is, what are the critical factors that influence phytoplankton production, which includes such aspects as photosynthesis, growth rates, and maintenance of biomass? Bottom-up processes refer to such processes as nutrient limitation (resource availability) and light limitation; top-down processes refer to predation (grazing). Consult the following citations and references therein for information on these concepts:

Lehman, J.T. 1991. Interacting growth and loss rates: the balance of top-down and bottom-up controls in plankton communities. *Limnol. Oceanogr.* 36: 1546–1554.

Lucas, L.V. 1998. Does the Sverdrup critical depth model explain bloom dynamics in estuaries? *J. Mar. Res.* 56: 375–415.

Wickham, S.A. 1995. Trophic relations between cyclopoid copepods and ciliated protists: complex interactions link the microbial and classical food webs. *Limnol. Oceanogr.* 40: 1173–1181.

VII. Primary Productivity of the Biosphere

This relatively brief section is not divided into subunits. It compares the relative contributions of oceanic net primary productivity with those of terrestrial systems. Before the use of satellite technology, the marine environment came out well on the short side of this comparison. Now we find that the two are quite comparable: Oceanic net primary productivity is approximately 49 petagrams of carbon, and terrestrial net primary productivity is about 56 petagrams of carbon. This volume provides other ways of looking at this comparison.

The Ocean Ecosystem

The next two long sections discuss different views of the oceanic ecosystem. The first is the traditional view of the world ocean: the classical model. The other is a newer view, one that is still being developed: a changing model. The former examines the theoretical underpinnings of ocean studies by looking at interactions among the larger plankton. The latter provides an overview of the directions that modern biological oceanography is taking in studying the more elusive and enigmatic microplankton.

VIII. The Ocean Ecosystem: The Classical Model

The classical model describes how the interactions of the larger plankton (net phytoplankton and copepods) control the oceanic ecosystem. This section is divided into two parts:

- Grazing
- Seasonal Succession in Phytoplankton

allelochemicals	diurnal migration	patchiness
allomone	kairomone	pheromones
biological conditioning	light-damage avoidance	red tide
<i>Calanus finmarchicus</i>	metabolic advantage	reverse migration
climax community	metabolites	seasonal succession
Copepoda	micrograzer hypothesis	toxin
copepodid	mixing hypothesis	UV light
coupling	naupliar larvae	visual predators
deep scattering layer	<i>Neocalanus cristatus</i>	vitamins
diel vertical migration (DVM)	<i>Neocalanus plumchrus</i>	
dinoflagellates	nocturnal migration	

1. Copepods can remove phytoplankton cells at a remarkable rate and can significantly reduce the phytoplankton population over a brief time span even if the photosynthetic rate is high. Intense grazing by copepods can account for the sudden decline of the spring bloom in phytoplankton production.
2. Copepods are dominant grazers in the world's oceans and thus significant enough in their own right to warrant a study of their life cycles. Unfortunately, few species have been completely studied. The life cycles of *Calanus finmarchicus* (North Atlantic Ocean) and *Neocalanus plumchrus* (North Pacific Ocean) have been studied in some detail.

There appears to be an efficient coupling of phytoplankton and zooplankton populations in the Pacific Ocean, but in the Atlantic Ocean the coupling is not as tightly connected. As a result, fixed carbon of the spring bloom in the Atlantic winds up in detritus. According to the classical model, the fate of carbon in these two ecosystems differs greatly. In the Atlantic, benthic fish ultimately receive the energy, presumably through benthic invertebrates, while in the Pacific, pelagic fish consume the copepod feast.

3. Diel vertical migration (DVM) refers to the daily up-and-down movements of zooplankton and other small organisms. This commuting is usually done such that the organisms are found at depth during daylight hours and in waters nearer the surface at night. Reverse migrations also are known to occur: daylight rise and midnight sinking. The how and the why of this up-and-down movement has been the subject of much debate since the phenomenon was first discovered more than 100 years ago. The text provides an excellent review of the various models that have been developed to explain both the proximal and ultimate causes of vertical migrations. Other resources include the following:

Bollens, S.M. 1996. Diel vertical migration in zooplankton. *Oceanus* 39: 19.

Dagg, M.J. 1997. Vertical migration and feeding behavior of *Calanus pacificus* females during a phytoplankton bloom in Dabob Bay, U.S. *Limnol. Oceanogr.* 42: 974–980.

Ohman, M.D. 1990. The demographic benefits of diel vertical migration by zooplankton. *Ecol. Monogr.* 60: 257–281.

High-resolution SONAR has been used to track the migration of krill in the Gulf of Maine. Apparently, these organisms accomplish a highly synchronous DVM, including a rapid ascent of some 50 m over the course of 15 min. For more information, consult the following reference:

Greene, C.H., et al. 1992. The migration behavior, fine structure, and bioluminescent activity of krill sound-scattering layers. *Limnol. Oceanogr.* 37: 650–658.

4. Composition of the phytoplankton in any given area changes with the season, with the pattern more or less repeating itself each year. This phenomenon is known as the seasonal succession of the phytoplankton, but it is not limited to marine ecosystems; it occurs in freshwater, too. For a classical limnologist's perspective, refer to the work of G. Evelyn Hutchinson, included in the following list of references.

Lindell, D., and A.F. Post. 1995. Ultraphytoplankton succession is triggered by deep winter mixing in the Gulf of Aqaba (Eilat), Red Sea. *Limnol. Oceanogr.* 40: 1130–1141.

Hutchinson, G.E. 1961. The paradox of the plankton. *Am. Nat.* 95: 137–146.

Hutchinson, G. E. 1967. Chapter 23 in *A treatise on limnology*, Volume II. New York: John Wiley & Sons.

IX. The Ocean Ecosystem: A Changing Model

While the classical model emphasizes interactions among the larger plankton, the changing model focuses on the photosynthetic activity of the nano- and picoplankton (coccolithophores, cyanobacteria, and prochlorophytes) and the fate of that production. Much of the information that was used to develop this model came from modern technology (such as remote sensing and flow cytometry). This section consists of six parts:

- Microorganisms and Productivity
- Nanoplankton, Respiration, and Grazing
- Bacteria, Particulate, and Dissolved Organic Matter
- The Marine Planktonic Food Web
- Spatial Distribution of Plankton
- The Major Plankton Biomes

¹⁴ C method	eddies	patches
advection	epifluorescence	picoplankton
anticyclonic	epipelagic zone	planktonic food web
bacterioplankton	eutrophic	POC
chlorophyll maximum	flow cytometry	prochlorophytes
chloroplast symbiosis	forams	Protista
coastal fronts	grazing	radiolarians
compensation depth	gyres	remote sensing
cyanobacteria	Langmuir cells	respiration
cyclonic	leaking cells	rings
differential interference contrast (DIC)	marine snow	satellite oceanography
dissolved organic carbon (DOC)	membrane filters	slope water
DOM	mesotrophic	symbionts
DON	microbial loop	symbiosis
DOP	nanoplankton	viruses
ecotones	oligotrophic	

1. The nanoplankton, particularly coccolithophores, probably account for much more of the primary productivity and biomass than was previously assumed. Nanoplankton may account for 80% or more of the photosynthetic activity in oceanic water. A temporary symbiosis, called chloroplast symbiosis, between amoeboid protozoans, such as forams and radiolarians, and various phytoplankton may account for an important fraction of primary production in the open seas. The primary consumers of nanoplankton are probably very small animal protists, perhaps nonphotosynthetic flagellates and ciliates.
2. Dissolved organic carbon (DOC) [or dissolved organic material (DOM)] in the oceans is the largest reserve of organic carbon in the biosphere. It has been estimated that up to 50% of carbon cycled through the DOC (from photosynthetically fixed carbon that was lost or leaked from cells) is taken up directly by bacterioplankton. At one time, DOC lost from algal cells was considered to have been lost from the planktonic community. However, it was found that this carbon is used by bacteria that are then eaten by nanozooplankton. Thus, a return loop is completed. This cycle, called the microbial loop, cycles carbon back into the planktonic community. Another important player in this loop are the viruses. Refer to Figures 2.46 and 2.48 in the textbook. The following references provide an interesting view:

Azam, F. 1998. Microbial control of ocean carbon flux: the plot thickens. *Science* 280: 694–696.

Balter, M. 2000. Evolution on life fringes. *Science* 289: 1866–1867. [N.B.: “Seagoing viruses . . . lock up at least 20 times more carbon than contained in all the whales on earth.”]

Mei, M.L., and R. Danovaro. 2004. Virus production and life strategies in aquatic sediments. *Limnol. Oceanogr.* 49: 459–470.

Wommack, K.E., and R.R. Colwell. 2000. Virioplankton: viruses in aquatic ecosystems. *Microbiol. Mol. Biol. Rev.* 64: 69–114.

3. Marine snow is a relatively new concept whose study requires the application of open-ocean SCUBA diving; nets simply are too coarse to retain the stuff, or they tear it into unrecognizable bits. Following are a few eclectic references to get students started:

Brzezinski, M.A. 1997. Silica cycling within marine snow. *Limnol. Oceanogr.* 42: 1706–1713.

Cox, V. 1994. It's no snow job: one woman's story of underwater discovery. *Sea Frontiers* 40: 42–49.

Kiorboe, T. 2000. Colonization of marine snow aggregates by invertebrate zooplankton: abundance, scaling and possible role. *Limnol. Oceanogr.* 45: 479–484.

MacIntyre, S., A.L. Alldredge, and C.C. Gotschalk. 1995. Accumulation of marine snow at the density discontinuities in the water column. *Limnol. Oceanogr.* 40: 449–468.

4. With the information presented above in mind, we must now revise the older view that large phytoplankton dominate primary production in the marine environment. We should not neglect these minute forms in our view of the oceans.
5. Phytoplankton tend to occur in patches of a scale varying from a few meters to hundreds of kilometers. These patches are produced by both abiotic and biotic forces. (See Fig. 4.49 in the textbook.) In spite of all the variables affecting the plankton that have been considered in this chapter, the general distribution of the plankton may be recognized as falling into specific, and reasonably well-demarcated, patterns.

ENRICHMENT EXERCISES

Sizes of Microbes

If you consult the International System of Measurement (SI) for assigning units, you will find disagreement with the prefixes used in Table 2.1 to describe some of the size categories of plankton. You may consult any recent edition of the *CRC Handbook of Chemistry and Physics* to see this fact for yourself, or, if your students are like mine, just wait for a chemistry major who has enrolled in your course to call you on this matter. The problem is that the SI units *micro* (10^{-6}), *nano* (10^{-9}), *pico* (10^{-12}), and *femto* (10^{-15}) do not correspond to those reported in the bottom row of Table 2.1. The synopsis of the development of these categories given in the next paragraph will help you explain this apparent inconsistency to your students.

Table 2.1 was modified slightly from Sieburth's (1979) Table 4-1. (See also the earlier version in Sieburth et al., 1978.) At the upper end of this formulation, Sieburth et al. (1978) start with the meter nekton (2–20 m, ca. 10^1 m), which is not shown in Table 2.1 in this text. The size scale below that is composed of a series of steps that does not exhibit an order-of-magnitude decrease in size at each step. Megaplankton, or decimeter nekton, are about 2–20 dm (ca. 10^{-1} to 10 m), and macroplankton, or centimeter nekton, are 2–20 cm (ca. 10^{-2} to 10^{-1} m). Except for the fact that the SI unit *mega* means 10^6 , these names make some sense, as does that of the next smallest unit, the mesoplankton (0.2–20 mm, ca. 10^{-4} to 10^{-2} m). The last four prefixes defined by Sieburth and his coworkers do not coincide with the SI units if one applies a linear measurement. However, Sieburth et al. (1978) argued that these units (such as micro) should be considered as volumetric (three dimensional). Thus, if the organisms are assumed to have a density approximating that of water (1 g/cm^3), then the prefix very roughly corresponds to the live weight of the organism.

“*Though this be madness, yet there is method in't.*”
Shakespeare (Polonius), *Hamlet*, 2:2:206.

Copley, J. 1999. Namibian bugs crash through the size barrier. *New Scientist* 162: 20.

Schulz, H.N. 1999. Dense populations of a giant sulfur bacterium in Namibian shelf sediments. *Science* 284: 493–495.

Sieburth, J.M. 1979. *Sea microbes*. New York: Oxford University Press.

Sieburth, J.M., V. Smetacek, and J. Lenz. 1978. Pelagic ecosystem structure: heterotrophic components of the plankton and their relationship to plankton size fractions. *Limnol. Oceanogr.* 23(6): 1256–1263.

Travis, J. 1999. Pearl-like bacteria are largest ever found. *Science News* 155: 246.

Variety in Microbes

For an excellent treatise of marine microbes, consult the textbook of Sieburth (1979) noted previously. This work covers prokaryotic and eukaryotic forms that have photosynthetic, chemolithotrophic, and heterotrophic metabolisms. Other interesting references and useful journals are noted as follows:

Austin, B. 1988. *Marine microbiology*. Cambridge, U.K.: Cambridge University Press.

Béjà, O., et al. 2000. Bacterial rhodopsin: evidence for a new type of phototrophy in the sea. *Science* 289: 1902–1906.

Cho, B.C., and F. Azam. 1992. Biogeochemical significance of bacterial biomass in the ocean's euphotic zone. *Mar. Ecol. Prog. Ser.* 63: 253–259.

Davies, D.G., et al. 1998. The involvement of cell-to-cell signals in the development of a bacterial biofilm. *Science* 280: 295–298.

Fenchel, T., G.M. King, and T.H. Blackburn. 1998. *Bacterial biogeochemistry: the ecophysiology of mineral cycling*. San Diego: Academic Press.

Fisk, M.R., S.J. Giovannoni, and I.H. Thorseth. 1998. Alteration of oceanic volcanic glass: textural evidence of microbial activity. *Science* 281: 978–980.

Fredrickson, J.K., and T.C. Onstott. 1996. Microbes deep inside the earth. *Sci. Am.* 175(4): 68–73.

Friedmann, E.I. (ed.). 1993. *Antarctic microbiology*. New York: John Wiley & Sons.

Guerrero, R., and L. Margulis. 1988. Stone soup. *The Sciences* 38(4): 34–38.

Jannasch, H.W., and M.J. Mottl. 1985. Geomicrobiology of deep-sea hydrothermal vents. *Science* 229: 717–725.

Kaiser, J. 1995. Can deep bacteria live on nothing but rocks and water? *Science* 270: 377.

Kolter, R., and R. Losick. 1998. One for all and all for one. *Science* 280: 226.

Rheinheimer, G. 1985. *Aquatic microbiology*, 3d ed. New York: John Wiley & Sons.

Simpson, S. 2000. Looking for life below. *Sci. Am.* 282(6): 94–101.

Skinner, F.A., and M.J. Shewan. (eds.). 1978. *Aquatic microbiology*. Society of Applied Bacteriology Symposia No. 6.

Stevens, T.O., and J.P. McKinley. 1995. Lithoautotrophic microbial ecosystems in deep basalt aquifers. *Science* 270: 450–454.

Also refer to the journals *Aquatic Microbial Ecology* and *Diseases of Aquatic Organisms*.

Shape in Radiolarians and Foraminiferans

The concepts of structure and function are recurring themes in biology (Thompson, 1971). In certain protozoans such as radiolarians, and, to some extent, foraminiferans, the structure is the geodesic dome. Chapter V in D'Arcy Thompson's classic book discusses this concept. This work also considers spiral shape in forams (Chapter VI). Its bibliographical information is as follows:

Thompson, D. 1971. Chapters V and VI in Bonner, J.T. (ed.). *On growth and form*. Cambridge, U.K.: Cambridge University Press.

Copepods

There is a rich literature on both freshwater and marine zooplankton suspension feeding. Researchers have monitored the decrease in suspended food particles with time, increase in radioactively labeled foods (bacteria, algae, and yeast) in the animals, and ingestion of various-sized latex microspheres (beads). The foods (for example, yeast or algae) are grown so that they incorporate a radioactive isotope in their tissues. As the animals feed, the concentrations of the label(s) increases in the organism. Latex microspheres are available from several sources (such as Sigma Chemical™ and Seradyn, Inc.™) and can be easily adapted into a variety of experimental procedures to determine clearance rates of zooplankton: mLs of water cleared of particles per animal per hour (mL/animal/hr). Simple manipulations of particle size, concentration, light, temperature, turbulence, or presence

or absence of predators or competitors will provide more than enough variations for an undergraduate lab exercise. Selected examples gleaned from a cursory examination of the literature are listed as follows, and more detailed searches will, no doubt, yield dozens of interesting papers:

- Alcaraz, M., G.-A. Paffenhöfer, and J.R. Strickler. 1980. Catching the algae: A first account of visual observations on filter feeding calanoids. Pages 241–248 in W.C. Kerfoot (ed.). *Evolution and ecology of zooplankton communities*. Hanover, NH: New England Press.
- Bundy, M.H., et al. 1993. Quantifying copepod searching efficiency using swimming pattern and perceptive ability. *Bull. Marine Sci.* 53: 15–28.
- Cowles, T.J., and J.R. Strickler. 1983. Characterization of feeding activity patterns in the planktonic copepod *Centropages typicus* Kroyer under various food conditions. *Limnol. Oceanogr.* 28: 106–115.
- Dam, H.G. 1993. Seasonal contrasts in the diel vertical distribution, feeding behavior, and grazing impact of the copepod *Temora longicornis* in Long Island Sound. *J. Mar. Res.* 51: 561–594.
- Harbison, G.R., V.L. McAlister, and R.W. Gilmer. 1986. The response of the salp, *Pegea confoederata*, to high levels of particulate material: starvation in the midst of plenty. *Limnol. Oceanogr.* 31: 371–382.
- Kerfoot, W.C., and K.L. Kirk. 1991. Degree of taste discrimination among suspension-feeding cladocerans and copepods: implications for detritivory and herbivory. *Limnol. Oceanogr.* 36: 1107–1123.
- Koehl, M.A.R., and J.R. Strickler. 1981. Copepod feeding currents: Food capture at low Reynolds number. *Limnol. Oceanogr.* 26: 1062–1073.
- Rothhaupt, K.O. 1990. Differences in particle size-dependent feeding efficiencies of closely related rotifer species. *Limnol. Oceanogr.* 35: 16–23.
- Strickler, J.R. 1982. Calanoid copepods, feeding currents, and the role of gravity. *Science* 218: 158–160.
- Vanderploeg, H.A., and G.-A. Paffenhöfer. 1985. Modes of algal capture by the freshwater copepod *Diaptomus sicilis* and their relation to food-size selection. *Limnol. Oceanogr.* 30: 871–885.
- Wylie, J.L., and D.J. Currie. 1991. The relative importance of bacteria and algae as food sources for crustacean zooplankton. *Limnol. Oceanogr.* 36: 708–728.
- Yen, J., et al. 1991. Feeding currents and energy dissipation by *Euchaeta rimana*, a subtropical pelagic copepod. *Limnol. Oceanogr.* 36: 362–369.

